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DBMS LAB 1-DDL COMMANDS

1.STUDENT TABLE:

```
CREATE TABLE Student (  
    student_id INT PRIMARY KEY,  
    student_name VARCHAR(100) NOT NULL,  
    email VARCHAR(100) UNIQUE  
);  
SHOW TABLES;  
ALTER TABLE Student ADD date_of_birth DATE;  
DESC Student;  
INSERT INTO Student (student_id, student_name, email, date_of_birth) VALUES  
(1, 'Alice Smith', 'alice@email.com', '2000-05-15'),  
(2, 'Bob Jones', 'bob@email.com', '1999-10-22'),  
(3, 'Charlie Brown', 'charlie@email.com', '2001-01-08');  
SELECT * FROM Student;  
DROP TABLE Student;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| student               |
+-----+

+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| student_id     | int           | NO   | PRI | NULL    |       |
| student_name   | varchar(100)  | NO   |     | NULL    |       |
| email          | varchar(100)  | YES  | UNI | NULL    |       |
| date_of_birth  | date          | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+

+-----+-----+-----+-----+-----+-----+
| student_id | student_name | email          | date_of_birth |
+-----+-----+-----+-----+-----+-----+
|          1 | Alice Smith  | alice@email.com | 2000-05-15    |
|          2 | Bob Jones   | bob@email.com   | 1999-10-22    |
|          3 | Charlie Brown | charlie@email.com | 2001-01-08    |
+-----+-----+-----+-----+-----+-----+
```

2.Course table:

```
CREATE TABLE Course (  
    course_code VARCHAR(10) PRIMARY KEY,  
    title VARCHAR(50) NOT NULL,  
    credits INT  
);  
SHOW TABLES;  
ALTER TABLE Course MODIFY title VARCHAR(150);  
DESC Course;  
INSERT INTO Course (course_code, title, credits) VALUES  
('CS101', 'Introduction to Computer Science', 4),  
('MATH201', 'Calculus I', 3),  
('ENG102', 'Literature and Composition', 3);  
SELECT * FROM Course;  
DROP TABLE Course;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| course                |
+-----+

+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| course_code   | varchar(10)   | NO   | PRI | NULL    |      |
| title         | varchar(150)  | YES  |     | NULL    |      |
| credits       | int           | YES  |     | NULL    |      |
+-----+-----+-----+-----+-----+-----+

+-----+-----+-----+-----+-----+-----+
| course_code | title                                | credits |
+-----+-----+-----+-----+-----+-----+
| CS101       | Introduction to Computer Science | 4       |
| ENG102      | Literature and Composition        | 3       |
| MATH201     | Calculus I                        | 3       |
+-----+-----+-----+-----+-----+-----+
```

3.Proffessor table:

```
CREATE TABLE Professor (  
    professor_id INT PRIMARY KEY,  
    first_name VARCHAR(50),  
    last_name VARCHAR(50),  
    contact_email VARCHAR(100)  
);  
SHOW TABLES;  
ALTER TABLE Professor DROP COLUMN contact_email;  
DESC Professor;  
INSERT INTO Professor (professor_id, first_name, last_name) VALUES  
(101, 'Alan', 'Turing'),  
(102, 'Marie', 'Curie'),  
(103, 'Albert', 'Einstein');  
SELECT * FROM Professor;  
DROP TABLE Professor;
```

Output:

+-----+						
Tables_in_sandbox_db						
+-----+						
professor						
+-----+						
+-----+-----+-----+-----+-----+-----+						
Field	Type	Null	Key	Default	Extra	
+-----+-----+-----+-----+-----+-----+						
professor_id	int	NO	PRI	NULL		
first_name	varchar(50)	YES		NULL		
last_name	varchar(50)	YES		NULL		
+-----+-----+-----+-----+-----+-----+						
+-----+-----+-----+						
professor_id	first_name	last_name				
+-----+-----+-----+						
	101	Alan	Turing			
	102	Marie	Curie			
	103	Albert	Einstein			
+-----+-----+-----+						

4.Department table:

```

CREATE TABLE Department (
    department_id INT PRIMARY KEY,
    dept_name VARCHAR(100),
    office_location VARCHAR(50)
);
SHOW TABLES;
ALTER TABLE Department RENAME COLUMN dept_name TO department_title;
DESC Department;
INSERT INTO Department (department_id, department_title, office_location) VALUES
(1, 'Computer Science', 'Building A'),
(2, 'Physics', 'Building B'),
(3, 'Mathematics', 'Building C');
SELECT * FROM Department;
DROP TABLE Department;

```

Output:

```

+-----+
| Tables_in_sandbox_db |
+-----+
| department            |
+-----+
+-----+-----+-----+-----+-----+-----+
| Field                | Type                | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| department_id        | int                 | NO   | PRI | NULL    |      |
| department_title     | varchar(100)        | YES  |     | NULL    |      |
| office_location      | varchar(50)         | YES  |     | NULL    |      |
+-----+-----+-----+-----+-----+-----+
+-----+-----+-----+-----+-----+-----+
| department_id | department_title | office_location |
+-----+-----+-----+-----+-----+
|          1 | Computer Science | Building A      |
|          2 | Physics          | Building B      |
|          3 | Mathematics      | Building C      |
+-----+-----+-----+-----+-----+

```

5. Classroom table:

```
CREATE TABLE Classroom (  
    room_number VARCHAR(10) PRIMARY KEY,  
    building VARCHAR(50),  
    capacity INT  
);  
SHOW TABLES;  
ALTER TABLE Classroom MODIFY building VARCHAR(100);  
DESC Classroom;  
INSERT INTO Classroom (room_number, building, capacity) VALUES  
('101A', 'Science Hall', 50),  
('204B', 'Arts Block', 30),  
('301C', 'Main Building', 100);  
SELECT * FROM Classroom;  
DROP TABLE Classroom;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| classroom              |
+-----+

+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| room_number | varchar(10)   | NO   | PRI | NULL    |      |
| building    | varchar(100)  | YES  |     | NULL    |      |
| capacity    | int           | YES  |     | NULL    |      |
+-----+-----+-----+-----+-----+-----+

+-----+-----+-----+
| room_number | building      | capacity |
+-----+-----+-----+
| 101A        | Science Hall  | 50       |
| 204B        | Arts Block    | 30       |
| 301C        | Main Building | 100      |
+-----+-----+-----+
```

6.Library table:

```
CREATE TABLE LibraryBook (  
    book_id INT PRIMARY KEY,  
    book_title VARCHAR(200) NOT NULL,  
    author_name VARCHAR(100)  
);  
SHOW TABLES;  
ALTER TABLE LibraryBook ADD isbn_number VARCHAR(20);  
DESC LibraryBook;  
INSERT INTO LibraryBook (book_id, book_title, author_name, isbn_number) VALUES  
(1, 'Database System Concepts', 'Silberschatz', '9780073523323'),  
(2, 'Clean Code', 'Robert C. Martin', '9780132350884'),  
(3, 'Design Patterns', 'Erich Gamma', '9780201633610');  
SELECT * FROM LibraryBook;  
DROP TABLE LibraryBook;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| librarybook          |
+-----+

+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| book_id        | int           | NO   | PRI | NULL     |       |
| book_title     | varchar(200)  | NO   |     | NULL     |       |
| author_name    | varchar(100)  | YES  |     | NULL     |       |
| isbn_number    | varchar(20)   | YES  |     | NULL     |       |
+-----+-----+-----+-----+-----+-----+

+-----+-----+-----+-----+-----+-----+
| book_id | book_title          | author_name          | isbn_number
+-----+-----+-----+-----+-----+-----+
|      1 | Database System Concepts | Silberschatz      | 9780073523323
|      2 | Clean Code              | Robert C. Martin  | 9780132350884
|      3 | Design Patterns         | Erich Gamma       | 9780201633610
+-----+-----+-----+-----+-----+-----+
```

7.University club:

```
CREATE TABLE UniversityClub (  
    club_id INT PRIMARY KEY,  
    club_name VARCHAR(100) NOT NULL,  
    budget DECIMAL(10, 2)  
);  
SHOW TABLES;  
ALTER TABLE UniversityClub RENAME TO StudentOrganization;  
DESC StudentOrganization;  
INSERT INTO StudentOrganization (club_id, club_name, budget) VALUES  
(1, 'Robotics Club', 5000.00),  
(2, 'Debate Society', 1500.50),  
(3, 'Photography Club', 2000.00);  
SELECT * FROM StudentOrganization;  
DROP TABLE StudentOrganization;
```

```

+-----+
| Tables_in_sandbox_db |
+-----+
| universityclub |
+-----+

+-----+-----+-----+-----+-----+-----+
| Field      | Type                | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| club_id    | int                 | NO   | PRI | NULL    |      |
| club_name  | varchar(100)        | NO   |     | NULL    |      |
| budget     | decimal(10,2)       | YES  |     | NULL    |      |
+-----+-----+-----+-----+-----+-----+

+-----+-----+-----+
| club_id | club_name      | budget |
+-----+-----+-----+
|      1 | Robotics Club  | 5000.00 |
|      2 | Debate Society | 1500.50 |
|      3 | Photography Club | 2000.00 |
+-----+-----+-----+

```